22 · Near vision: the three changes

The words of this track, as it says them. Bold brown is the examiner.

Near vision. Three marks.

What changes occur in the eye to allow near vision? And how is this undone for distant vision?

Three marks, three changes. Picture yourself reading a book held close to your face.

The hook: bulge, shrink, cross. The lens bulges, the pupils shrink, and the eyes cross inwards. Look out of the window, and all three undo.

Bulge. The ciliary muscle contracts, the suspensory ligaments go slack, and the elastic lens recoils into a rounder shape, so it bends light more. That's accommodation of the lens.

Shrink. The pupils constrict. The circular, sphincter muscles of the iris contract.

Cross. The eyes converge: both eyeballs rotate medially, turning inwards toward the near object.

Why it's needed. Light from a near object arrives as diverging rays. Light from more than six metres away arrives as parallel rays. The near rays need more bending to land on the retina.

For distant vision, all three undo. The ciliary muscle relaxes, the lens flattens, the pupils dilate, and the eyes look straight ahead.

The traps. For near vision, the pupil constricts. Her student-marked feedback says it dilates, but her own quiz key and both of her decks say constricts. Write constricts. Second: the ciliary muscle contracts to make the lens bulge. Her true or false trap says it contracts to flatten the lens. That's false. Third: one of her slides calls the lens biconcave. It's biconvex.

Once more. Near vision.

The ciliary muscle contracts, so the lens bulges and gets rounder: accommodation. The pupils constrict. The eyes converge, turning inwards. For distant vision, all three undo: the ciliary muscle relaxes, the lens flattens, and the pupils dilate. Near light arrives diverging; from over six metres away it's parallel.